QUANT SINGULARITY

Financial ML — Intern Screening Project

Summer 2026 intake · Financial ML track

Context

Quant Singularity is building an AI-supervised systematic trading platform for Indian derivatives markets. The Financial ML track owns a core part of the stack: feature engineering, model training, simulation, and the post-trade learning loop that makes our system continuously improve.

This project is the final screening stage for the Financial ML intern role. You have already cleared the psychometric and technical assessments. The brief below is designed to show us how you think, not only how you code. We care far more about the rigor of your analysis than the headline Sharpe ratio of your model.

Read the entire document before you start. Several design choices are explicitly left to you and will be assessed on their defensibility.

The project

Build a direction-prediction classifier for NIFTY 50, with full end-to-end validation. The model itself is not the point. What matters is everything around it: data handling, validation discipline, feature design, and the intellectual honesty of your reporting.

Specification

Parameter	Specification
Target	NIFTY 50 next-day close direction (binomial: up = 1, down = 0)
Information cutoff	End of day T; prediction is for day T+1
Timeline	4 calendar days from brief release. Plan for 15 to 20 hours of work.
In-sample window	January 2022 to June 2025
Out-of-sample window	July 2025 to December 2025 (held back; do not tune on this)

Parameter	Specification
Model requirement	One gradient boosting model (XGBoost or LightGBM). A second model is optional and will not add marks if the first is weak.
Feature cap	12 features in the final submission
Validation	Walk-forward. Standard k-fold is disqualifying.
Experiment tracking	MLflow, from the first run. Retrofitted tracking will be detected.
Report length	4 to 6 pages as PDF

Data we provide

You will receive a single zip containing:

- NIFTY 50 daily OHLCV for January 2022 to December 2025 (full window, including the held-back period you must not tune on).
- Bank Nifty daily OHLCV over the same window, as a permitted cross-asset input.
- India VIX daily close over the same window.
- A pre-built starter feature file (starter_features.csv) containing approximately 30 candidate features.

A deliberate note on the starter features: we have not documented exactly how this file was built. Treat it as you would any dataset arriving from a real third-party data vendor. Not all features in the file are safe to use as provided. Part of your job is to figure out which ones you trust.

Data you may not use

To keep submissions comparable, do not add any of the following for this exercise: intraday data, options chain data, news or sentiment, fundamentals, alternative data (FII/DII flows, open interest, etc.), or data from periods outside the provided window. If you would want one of these in a real setting, mention it in the appendix with a short justification, but do not incorporate it into the model.

Deliverables

Submit three things, in this order of importance:

- **Written report.** PDF, 4 to 6 pages, structured per the section prompts below. This is the most heavily weighted artifact.
- **Code repository.** Private GitHub repo containing your code, MLflow artifacts, requirements.txt (or pyproject.toml), and a README with setup plus run instructions.
- **Live walkthrough.** A 10-minute session with us in the final round. We will ask you to explain and modify your code in real time.

The report: structure and prompts

The report carries more weight than the code. A clean model paired with a thoughtful report will outscore a brilliant model with a thin writeup. Address every section below.

1. Problem framing (around 1 page)

What are you predicting, at what horizon, using what information? How do you treat flat or near-zero returns — up bucket, down bucket, dropped? Why? What is your baseline (majority class, random, persistence) and what accuracy does it achieve? Your model must beat these baselines meaningfully to be interesting.

2. Data audit (around 1 page)

What did you find in the provided dataset? Any issues, inconsistencies, or concerns? Any features you chose to drop, and why? Anything about the starter feature file that gave you pause? Be specific. Vague data audits score poorly.

3. Feature engineering and EDA (around 1 page)

List your final 12 features. For each, give a one-line justification and state the exact timestamp at which that feature is safely computable. Summarise EDA briefly: correlation structure, multicollinearity handling, scaling decisions. Do not reproduce many plots — show the two or three that actually changed a decision you made.

4. Model and training (around half a page)

Architecture and hyperparameter search strategy. Walk-forward split scheme described precisely: window sizes, step sizes, refit cadence. Why you made the choices you made. We want the reasoning, not just the final config.

5. Results (around 1 page)

On the out-of-sample window, report: AUC, balanced accuracy, confusion matrix, and trading-strategy backtest metrics (Sharpe, max drawdown, hit rate, turnover). Confidence intervals on every metric, not point estimates. Compare to your baselines from Section 1.

6. How do I know this edge is real? (at least 1 page — critical)

This is the section we care about most. Address, at minimum:

- Statistical significance of your out-of-sample edge.
- Sensitivity of results to the train/test split boundaries.
- Whether your results survive a same-distribution random-label baseline.
- Specific ways your backtest might still be lying to you.
- What you would build or test next if given another week, and what would break this approach first in live trading.

A candidate who writes *"Sharpe is 1.8, therefore the strategy works"* will not advance. A candidate who writes *"Sharpe is 1.8, but I am not yet confident, here is why, and here is what I would do next"* is exactly who we want to work with.

Evaluation rubric

We score on five dimensions. Weights are explicit and final.

Dimension	Weight	What we are assessing
Rigor of validation	**30%**	Walk-forward correctness. No lookahead leakage. Proper baselines. Honest uncertainty reporting.
Analysis and writeup	**25%**	Depth of Section 6. Clarity of reasoning. Honest acknowledgment of weaknesses and limits.
Engineering quality	**15%**	Code modularity. Reproducibility. MLflow discipline. Sensible structure and naming.
Feature thoughtfulness	**15%**	Not the count of features. The thinking behind each choice and how it was defended.
Model performance	**15%**	Out-of-sample metrics. Note this is the smallest weight, not the largest.

Note on the weightings: model performance is 15%, not 50%. We can teach modeling. We cannot easily teach the instincts that drive Sections 2 and 6. A mediocre model with an outstanding analysis will beat a great model with a weak one, every time.

Rules and constraints

- **Open-source libraries.** You may use any open-source library. Document every external dependency in your report.
- **LLM assistance.** You may use ChatGPT, Claude, Copilot, Cursor, or any other tool for coding help. We expect this. However, in the live walkthrough we will ask you to navigate and modify your own code — if you cannot, that is a strong negative signal regardless of how polished the submission looks.
- **Independent work.** You may discuss the problem conceptually with peers, but all code and writeup must be entirely your own.
- **No external data.** Do not use any data beyond what we provide. Appendix proposals for additional data are welcome but must not touch the model.
- **Walk-forward only.** Standard k-fold cross-validation on time-series data is disqualifying.
- **No tuning on the held-back window.** The July to December 2025 period is for final evaluation only. Any evidence of hyperparameter tuning against it will lead to disqualification.

Submission

Email a single link to your private GitHub repository (containing code, MLflow artifacts, README, and the PDF report) to the address provided in your invitation.

Deadline: as stated in your invitation email, in IST. Late submissions will not be reviewed. Do not compress the repository into a zip unless GitHub rejects specific files, in which case note this in the README.

A final note

The ambiguities and omissions in this brief are intentional. We are not trying to catch you out for sport. We are trying to see how you handle a realistic quant research setup, which is never cleanly specified. A careful, honest submission that flags an issue with the data and explains how you handled it will always outscore a high-Sharpe submission that silently used contaminated inputs.

We are looking for someone who will own a component of a live trading system in a few weeks. Write accordingly.

Direct your questions to: surya@quantsingularity.in Good luck. We look forward to reading what you build.

Surya
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